IGCSE 0607 Practice Questions — E5.2 — Combined Questions & Markscheme

Angle measurement in degrees

Total Questions: 24 — **Total Marks:** 144 Marks

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- **E5.2** Angle measurement in degrees
  → **Paper 2 (Non-Calculator)** *(12 Qs)* · **Paper 4 (Calculator)** *(12 Qs)*

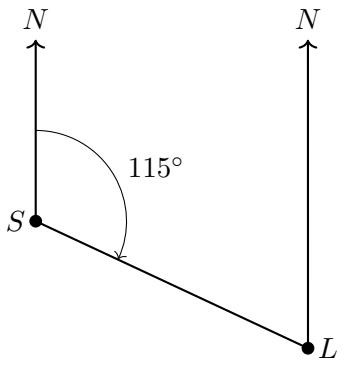
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1 The bearing of point Y from point X is 048°.

Work out the bearing of point X from point Y.

..... ° **[2]**

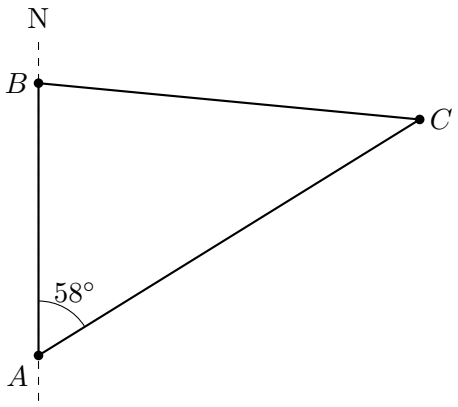
2 The diagram shows the positions of a ship, S, and a lighthouse, L.



The bearing of the lighthouse from the ship is 115°.
Work out the bearing of the ship from the lighthouse.

..... ° **[2]**

3 The diagram shows the positions of three towns, A, B and C.
B is due north of A and the bearing of C from A is 058°.



(a) Find the bearing of A from C.

..... ° **[2]**

(b) Given that angle $ABC = 84°$, find the bearing of C from B.

..... ° **[1]**

4 The positions of three towns, A, B and C, are marked on a map.

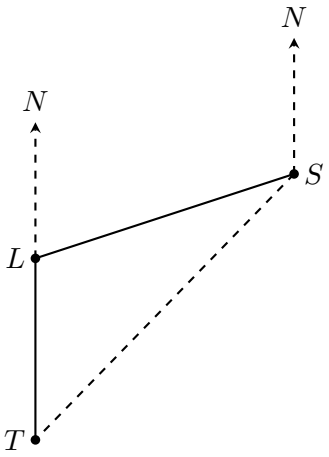
(a) The bearing of town B from town A is 068°.
Work out the bearing of town A from town B.

..... ° **[2]**

(b) Town C is due west of town B.
Write down the bearing of town C from town B.

..... ° **[1]**

5 The diagram shows the positions of a lighthouse L and two ships S and T.
The bearing of ship S from the lighthouse L is 072°.



(a) Calculate the three-figure bearing of the lighthouse L from ship S.

..... ° **[2]**

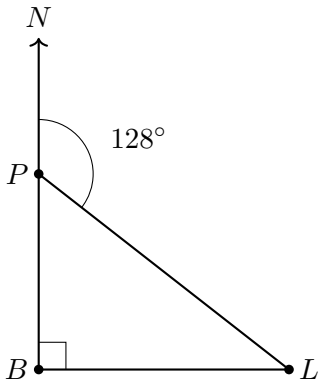
(b) Ship T is due South of the lighthouse L.
Calculate the size of angle SLT.

$\angle SLT =$ ..... ° **[1]**

(c) The bearing of ship T from ship S is 215°.
Calculate the size of angle LST.

$\angle LST =$ ..... ° **[2]**

6 The bearing of a lighthouse L from a port P is 128°.
A boat B is positioned such that it is due South of P and due West of L.



(a) Find the bearing of P from L.

..... ° **[2]**

(b) Work out the size of angle BPL.

..... ° **[1]**

(c) Work out the size of angle PLB.

..... ° **[2]**

7 The bearing of a lighthouse, L, from a boat, B, is 148°.

(a) Work out the bearing of B from L.

..... ° **[2]**

(b) A port, P, is due north of L and due east of B.

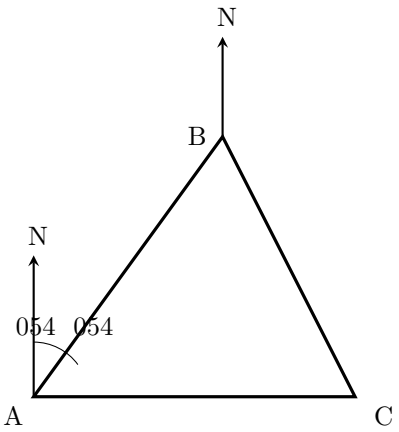
(i) Write down the bearing of B from P.

..... ° **[1]**

(ii) Work out the size of angle LBP.

..... ° **[2]**

8 The diagram shows the positions of three lighthouses,
A,
B, and
C. Lighthouse
C is due east of lighthouse
A, and the bearing of
B from
A is
054
054
.



(a) Write down the three-figure bearing of
C from
A.

.....

[1]

(b) Calculate the three-figure bearing of
A from
B.

.....

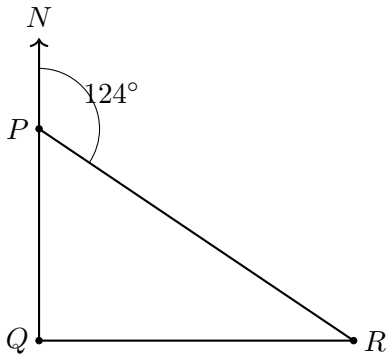
[2]

(c) Given that
=
AB=AC, calculate the three-figure bearing of
C from
B.

.....

[2]

9 The diagram shows the positions of three points, P, Q and R.
Point Q is due south of point P.
Point R is due east of point Q.
The bearing of point R from point P is 124°.



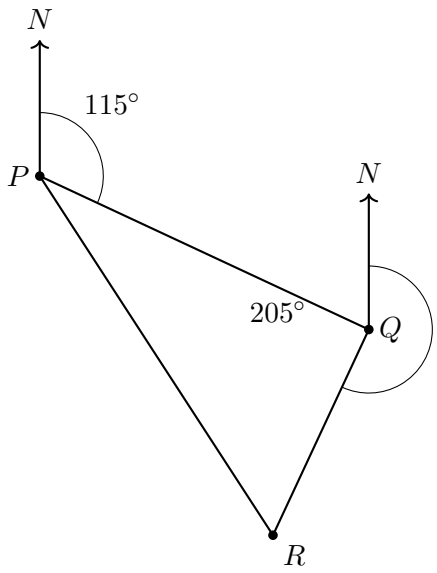
(a) Find the bearing of point P from point R.

..... ° **[2]**

(b) Calculate the angle PRQ.

Angle PRQ = ... ° **[2]**

10 The diagram shows the positions of three points P, Q and R.
The bearing of Q from P is 115°.
The bearing of R from Q is 205°.



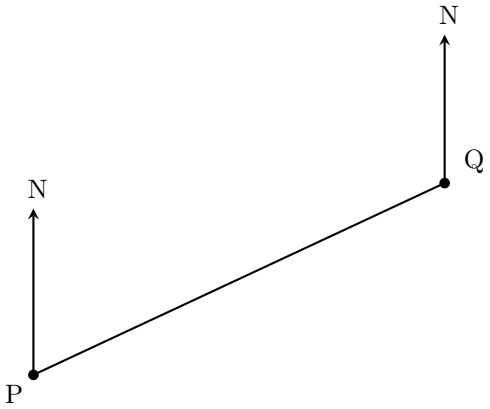
(a) Show that angle $PQR = 90°$.

[2]

(b) The distance PQ is 15 km and the distance QR is 8 km.
Calculate the distance PR.

..... km **[3]**

11 The scale drawing shows the positions of two towns,
P and
Q, on a map.
The scale is
1
1 cm represents
5
5 km.



(a) Measure the three-figure bearing of
Q from
P.

.....

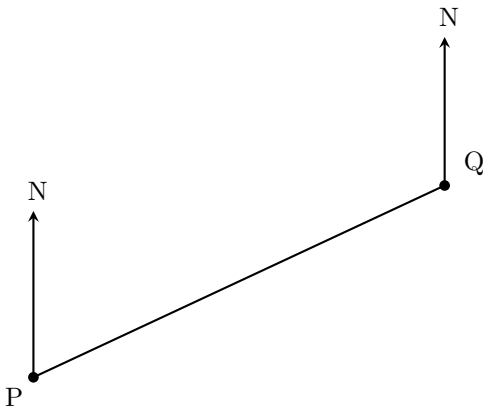
[1]

(b) Work out the actual distance, in kilometres, from town
P to town
Q.

..... km **[2]**

(c) A third town,
R, is
25
25 km from
Q on a bearing of
140
140
.
Using a ruler and protractor, mark and label the position of town

R on the scale drawing below.



[2]

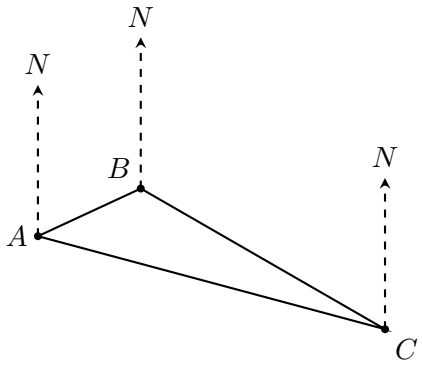
(d) Without measuring, calculate the three-figure bearing of P from Q.

.....

[2]

12 The diagram shows the positions of three points, A, B, and C.

- The bearing of B from A is 065°.
- The bearing of C from B is 120°.
- The bearing of A from C is 285°.



(a) Calculate the size of angle ABC.

$\angle ABC =$ ..... ° **[2]**

(b) Calculate the size of angle BCA.

$\angle BCA =$ ..... ° **[2]**

(c) Point D lies on the straight line AC such that triangle ABD is isosceles with $AB = BD$.

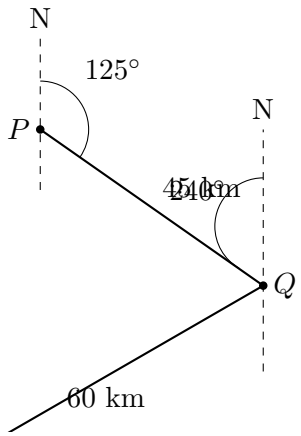
(i) Calculate the three-figure bearing of C from A.

..... ° **[2]**

(ii) Calculate the size of angle ABD.

$\angle ABD =$ ..... ° **[2]**

13 A ship sails from port P on a bearing of 125° for 45 km to point Q.
It then sails from Q on a bearing of 240° for 60 km to point R.



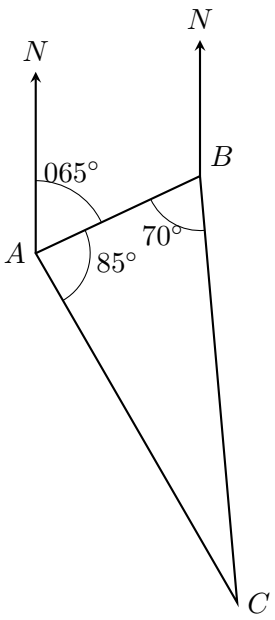
(a) Show that angle $PQR = 65°$.

[2]

(b) Calculate the distance PR.

$PR =$ ..... km **[3]**

14 The diagram shows the positions of three towns, A, B and C.
The bearing of B from A is 065°.
Angle $CAB = 85°$ and angle $ABC = 70°$.



(a) Calculate the bearing of C from A.

..... ° **[1]**

(b) Calculate the bearing of A from B.

..... ° **[2]**

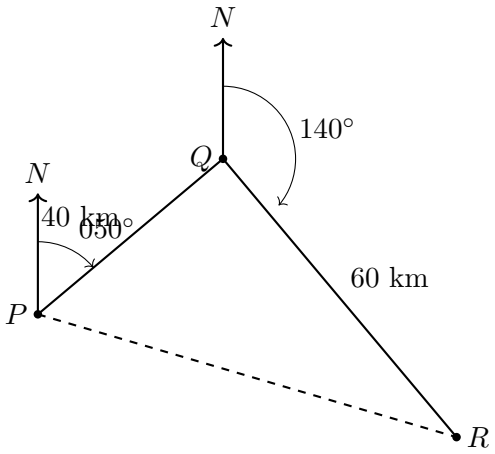
(c) Calculate the bearing of C from B.

..... ° **[2]**

(d) Calculate angle ACB.

..... ° **[1]**

15 A ship sails 40 km from a port P to point Q on a bearing of 050°.
It then sails 60 km from Q to point R on a bearing of 140°.



(a) Show that the angle PQR is 90°.

[2]

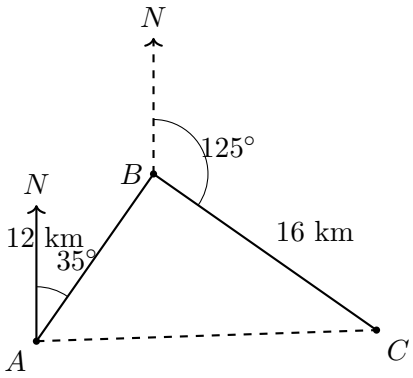
(b) Calculate the distance PR.

..... km **[2]**

(c) Calculate the bearing of R from P.

..... ° **[3]**

16 A ship leaves a port A and travels 12 km on a bearing of 035° to point B.
From B, the ship travels 16 km on a bearing of 125° to point C.



(a) Show that angle $ABC = 90°$.

[2]

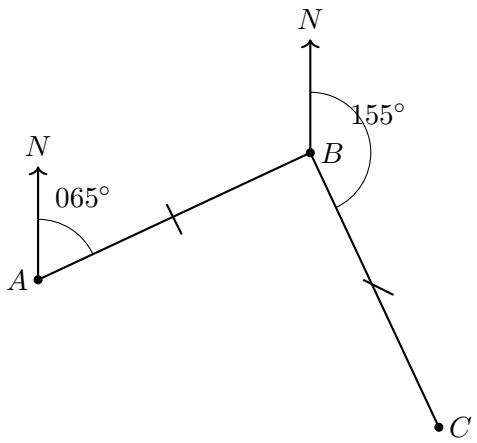
(b) Calculate the direct distance AC.

$AC =$ ..... km **[3]**

(c) Calculate the bearing of C from A.

..... ° **[3]**

17 Point B is on a bearing of 065° from point A.
Point C is on a bearing of 155° from point B.
The distance $AB = BC$.



(a) Show that angle $ABC = 90°$.

[2]

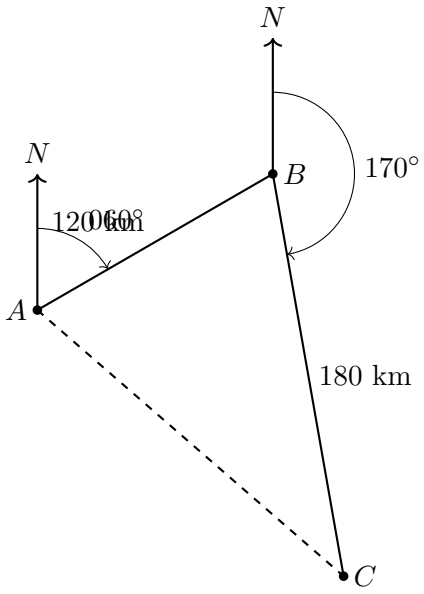
(b) Work out the bearing of A from C.

..... ° **[3]**

(c) Point D is such that the quadrilateral $ABCD$ is a square.
Find the bearing of D from A.

..... ° **[2]**

18 A helicopter flies from base A to base B, a distance of 120 km on a bearing of 060°.
It then flies from base B to base C, a distance of 180 km on a bearing of 170°.



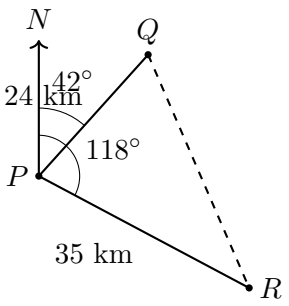
(a) Calculate the distance AC.

..... km **[4]**

(b) Calculate the bearing of C from A.

..... ° **[3]**

19 Three towns are located at points P, Q, and R.
Town Q is 24 km from P on a bearing of 042°.
Town R is 35 km from P on a bearing of 118°.



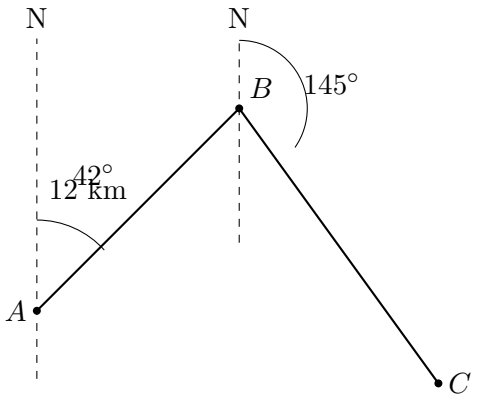
(a) Calculate the distance QR.

$QR =$ ..... km **[3]**

(b) Calculate the bearing of R from Q.

..... ° **[4]**

20 Three lighthouses A, B and C are situated such that B is 12 km from A on a bearing of 042°. C is 18 km from A and the bearing of C from B is 145°.



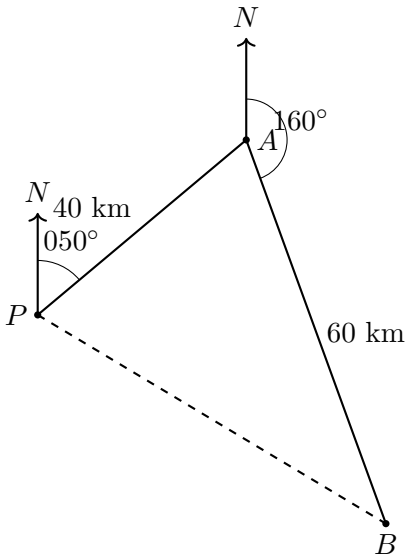
(a) Calculate the bearing of C from A.

..... ° **[5]**

(b) A boat sails from A in a straight line such that it is always equidistant from B and C. Calculate the bearing on which the boat sails.

..... ° **[2]**

21 A ship leaves port P and sails 40 km on a bearing of 050° to reach point A.
It then sails 60 km on a bearing of 160° to reach point B.



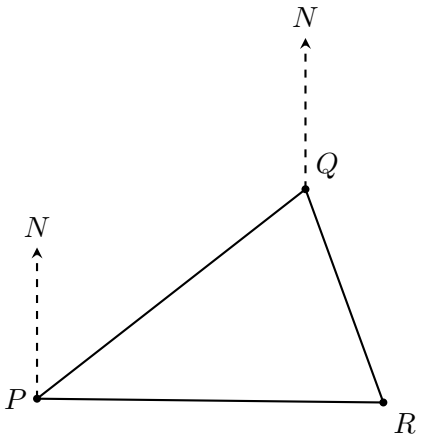
(a) Calculate the distance PB.

..... km **[4]**

(b) Calculate the bearing of B from P.

..... ° **[4]**

22 The points P, Q, and R represent the positions of three towns.
Town Q is 45 km from town P on a bearing of 052°.
Town R is 30 km from town Q on a bearing of 160°.



(a) Show that the angle PQR is 72°.

[2]

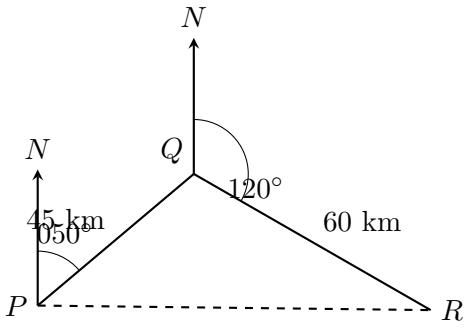
(b) Calculate the distance PR.

$PR =$ ..... km **[3]**

(c) Calculate the three-figure bearing of town R from town P.

..... ° **[4]**

23 A ship sails 45 km from Port P on a bearing of 050° to Point Q.
It then sails 60 km on a bearing of 120° to Point R.



(a) Calculate the distance PR.

..... km **[4]**

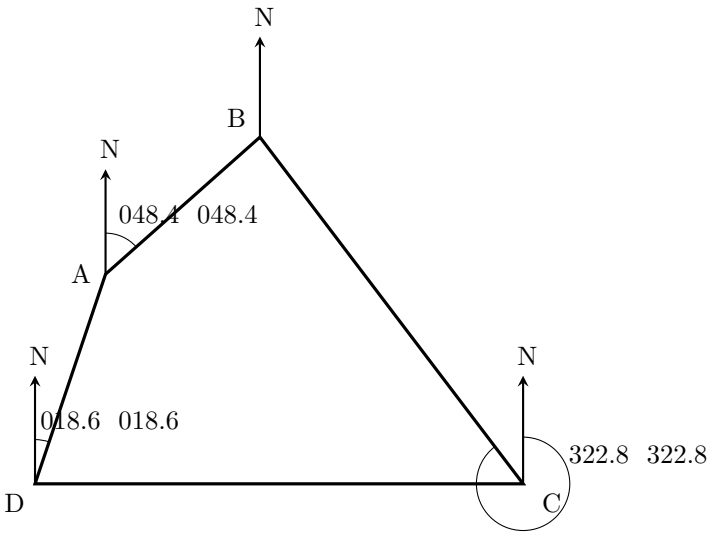
(b) Calculate the bearing of R from P.

..... ° **[3]**

(c) The ship sails directly back from R to P.
Calculate the bearing of P from R.

..... ° **[2]**

24 The diagram shows the positions of four radar stations,
A,
B,
C, and
D, on flat ground.
Station
C is due east of station
D. The bearing of
A from
D is
018.6
018.6
, the bearing of
B from
A is
048.4
048.4
, and the bearing of
B from
C is
322.8
322.8
.



(a) Write down the three-figure bearing of
D from
C.

.....

[1]

(b) Calculate the three-figure bearing of
D from
A.

. . . . . .

[2]

(c) Calculate the size of

(i) angle DAB,

Angle = DAB= . . . . . .

[2]

(ii) angle ABC.

Angle = ABC= . . . . . .

[3]

(d) Calculate the three-figure bearing of C from B.

. . . . . .

[2]

1 $048 + 180$ **[M1]**

228 **[A1]**

[Back to Question] *[Total Marks: 2]*

2 Identifies the relationship between alternate angles or uses $180° + 115°$ **[M1]**

295 **[A1]**

[Back to Question] *[Total Marks: 2]*

3 (a) $180 + 58$ **[M1]**

238 **[A1]**

(b) $180 - 84 = 096$ or 096 **[B1]**

[Back to Question] *[Total Marks: 3]*

4 (a) $068 + 180$ **[M1]**

248 **[A1]**

(b) 270 **[B1]**

[Back to Question] *[Total Marks: 3]*

5 (a) $072° + 180°$ **[M1]**

252 **[A1]**

(b) $180° - 72° = 108°$ **[B1]**

(c) Identifies the bearing of L from S is $252°$ **[M1]**

$252° - 215° = 37$ **[A1]**

[Back to Question] *[Total Marks: 5]*

6 (a) $128 + 180$ **[M1]**

308 **[A1]**

(b) 52 **[B1]**

(c) $180 - 90 - 52$ **[M1]**

38 **[A1]**

[Back to Question] *[Total Marks: 5]*

7 (a) $180 + 148$ OR $360 - (180 - 148)$ **[M1]**

328 **[A1]**

(b) **(i)** 270 **[B1]**

(ii) Identifies the bearing of P from B is 090° **[B1]**

$148 - 90 = 58$ **[A1]**

[Back to Question] *[Total Marks: 5]*

8 **(a)**

090
090 **[B1]**

(b)

054
+
180
054
+180

[M1]

234
234 **[A1]**

(c)

Angle
=
90
54
=
36
BAC=90
54
=36
and angle
=
180
36
2
=
72
ABC=
2
180
36
=72

[M1]

Bearing
=
234
72
=
162
=234
72
=162 **[A1]**

[Back to Question] *[Total Marks: 5]*

9 **(a)** $124 + 180$ **[M1]**

304 **[A1]**

(b) $\angle PQR = 90°$ or $\angle QPR = 180° - 124° = 56°$ **[M1]**

$\angle PRQ = 180° - 90° - 56° = 34°$ **[A1]**

[Back to Question] *[Total Marks: 4]*

10 **(a)** Bearing of P from $Q = 115 + 180 = 295°$ **[M1]**

Angle $PQR = 295° - 205° = 90°$ **[A1]**

(b) $15^2 + 8^2$ **[M1]**

$225 + 64 = 289$ **[M1]**

17 **[A1]**

[Back to Question] *[Total Marks: 5]*

11 **(a)** 065
065 (accept
063
063 to
067
067) **[B1]**

(b) Measured length
=
6
PQ=6 cm (accept
5.8
5.8 cm to
6.2
6.2 cm) **[B1]**

30
30 (accept
29
29 to
31
31) **[A1]**

(c) Line drawn from
Q on bearing
140
140
(accept
138
138
to
142
142
) **[B1]**

Point
R marked at distance
5
5 cm from
Q (accept
4.8
4.8 cm to
5.2
5.2 cm) and labelled **[B1]**

(d) 065
\+
180
065
+180

[M1]

245
245 **[A1]**

[Back to Question] *[Total Marks: 7]*

12 **(a)** Calculates reverse bearing of A from $B = 065° + 180° = 245°$ **[M1]**

$245° - 120° = 125$ **[A1]**

(b) Calculates reverse bearing of B from $C = 120° + 180° = 300°$ **[M1]**

$300° - 285° = 15$ **[A1]**

(c) **(i)** $285° - 180°$ **[M1]**

105 **[A1]**

(ii) Calculates $\angle CAB = 105° - 065° = 40°$ (or $180° - 125° - 15° = 40°$) **[M1]**

$\angle ABD = 180° - 2(40°) = 100$ **[A1]**

[Back to Question] *[Total Marks: 8]*

13 **(a)** Interior angle at Q on North line $= 180° - 125° = 55°$ **[M1]**

Angle $PQR = 360° - 240° - 55° = 65°$ **[A1]**

(b) $PR^2 = 45° + 60° - 2(45)(60)\cos(65°)$ **[M2]**

$PR = 57.8$ or $57.82...$ **[A1]**

[Back to Question] *[Total Marks: 5]*

14 **(a)** 150 **[B1]**

(b) $180 + 65$ OR $360 - (180 - 65)$ **[M1]**

245 **[A1]**

(c) Identifies bearing of A from B is 245° (can follow through from (b)) **[M1]**

$245 - 70 = 175$ **[A1]**

(d) 25 **[B1]**

[Back to Question] *[Total Marks: 6]*

15 **(a)** Calculates the interior angle at Q on the North side: $180° - 50° = 130°$ OR bearing of P from $Q = 180° + 50° = 230°$ **[M1]**

Calculates angle PQR: $360° - 130° - 140° = 90°$ OR $230° - 140° = 90°$ **[A1]**

(b) Uses Pythagoras' theorem: $PR^2 = 40^2 + 60^2$ **[M1]**

$72.11\cdots = 72.1$ **[A1]**

(c) Uses trigonometry to find angle QPR: $\tan(QPR) = \dfrac{60}{40}$ **[M1]**

Angle $QPR = 56.3\ldots°$ **[M1]**

Bearing $= 050° + 56.3° = 106.3°$ **[A1]**

[Back to Question] *[Total Marks: 7]*

16 **(a)** Interior angle at B (South to BA) $= 35°$ OR interior angle from North line $= 180 - 35 = 145°$ **[M1]**

Angle $ABC = 360 - 145 - 125 = 90°$ OR $125 - 35 = 90°$ **[A1]**

(b) $AC^2 = 12^2 + 16^2$ **[M1]**

$AC^2 = 144 + 256 = 400$ **[M1]**

$AC = 20$ **[A1]**

(c) $\tan(\angle BAC) = \dfrac{16}{12}$ or $\sin(\angle BAC) = \dfrac{16}{20}$ **[M1]**

$\angle BAC = 53.13°$ **[A1]**

Bearing $= 035° + 53.13° = 088.1°$ **[A1]**

[Back to Question] *[Total Marks: 8]*

17 **(a)** Bearing of A from B is $065° + 180° = 245°$ or Interior angle between North and AB is $180° - 65° = 115°$ **[M1]**

$245° - 155° = 90°$ or $360° - 115° - 155° = 90°$ **[A1]**

(b) Triangle ABC is right-angled and isosceles, so $\angle BCA = (180° - 90°) \div 2 = 45°$ **[M1]**

Bearing of B from C $= 155° + 180° = 335°$ **[M1]**

Bearing of A from C $= 335° - 45° = 290°$ **[A1]**

(c) $\angle DAB = 90°$ **[M1]**

Bearing of D from A $= 065° + 90° = 155°$ **[A1]**

[Back to Question] *[Total Marks: 7]*

18 **(a)** Calculates angle $ABC = 240° - 170° = 70°$ (or equivalent geometric reasoning) **[B1]**

Uses the cosine rule: $AC^2 = 120° + 180^2 - 2(120)(180)\cos(70°)$ **[M1]**

$AC^2 = 14400 + 32400 - 43200(0.3420\ldots) = 32024.7\ldots$ **[M1]**

$178.9\cdots = 179$ (to 3 sf) **[A1]**

(b) Uses the sine rule to find angle BAC: $\dfrac{\sin(BAC)}{180} = \dfrac{\sin(70°)}{178.95}$ **[M1]**

$\sin(BAC) = 0.9452\cdots \implies$ Angle $BAC = 70.9°$ **[M1]**

Bearing $= 060° + 70.9° = 130.9°$ **[A1]**

[Back to Question] *[Total Marks: 7]*

19 **(a)** $\angle QPR = 118° - 42° = 76°$ **[M1]**

$QR^2 = 24^2 + 35^2 - 2(24)(35)\cos(76°)$ **[M1]**

$QR = 37.3$ or $37.34\ldots$ **[A1]**

(b) $\dfrac{\sin(\angle PQR)}{35} = \dfrac{\sin(76°)}{37.34}$ OR $\cos(\angle PQR) = \dfrac{24^2 + 37.34° - 35^2}{2(24)(37.34)}$ **[M1]**

$\sin(\angle PQR) = 0.909\ldots$ OR $\cos(\angle PQR) = 0.415\ldots$ **[M1]**

$\angle PQR = 65.4°$ or $65.42\ldots°$ **[A1]**

Bearing $= 180 + 42 - 65.4 = 156.6°$ (or $157°$) **[A1]**

[Back to Question] *[Total Marks: 7]*

20 **(a)** Angle $ABC = 360° - 145° - (180° - 42°) = 77°$ **[M1]**

$\dfrac{\sin(BAC)}{BC} = \dfrac{\sin(77°)}{18}$ **[M1]**

$BC^2 = 12^2 + 18^2 - 2(12)(18)\cos(BAC)$ (Implicitly requires finding BC or BAC first. Correct approach: $\dfrac{\sin(BCA)}{12} = \dfrac{\sin(77°)}{18}$) **[M1]**

$\sin(BCA) = \dfrac{12\sin(77°)}{18} = 0.6495...$ **[A1]**

Angle $BCA = 40.5°$ **[A1]**

Angle $BAC = 180° - 77° - 40.5° = 62.5°$ **[M1]**

Bearing $= 042° + 62.5° = 104.5°$ **[A1]**

(b) Boat sails on perpendicular bisector of BC. Angle $BAC = 62.5°$. The bisector of angle BAC is not necessarily the perpendicular bisector of BC.

Line is perpendicular bisector of BC.

Let midpoint of BC be M. In $\triangle ABM$, AM is the path.

Use cosine rule to find BC: $BC^2 = 12^2 + 18^2 - 2(12)(18)\cos(62.5°)$

BC = 16.39...

$\sin(ABM)/AM = \sin(BAM)/8.19...$ (This approach is overly complex).

Simpler: Path is perpendicular bisector. Angle of BC with North is 145°.

Perpendicular to BC has bearing $145° - 90° = 055°$.

Recognise path is perpendicular to BC **[M1]**

Bearing $= 145 - 90 = 055$ (or 055°) **[A1]**

[Back to Question] *[Total Marks: 7]*

21 (a) Bearing of P from $A = 050 + 180 = 230°$ **[M1]**

Angle $PAB = 230° - 160° = 70°$ **[M1]**

$PB^2 = 40^2 + 60^2 - 2(40)(60)\cos(70°)$ **[M1]**

59.6 or 59.65... **[A1]**

(b) Uses sine rule or cosine rule to find angle APB **[M1]**

$\frac{\sin(APB)}{60} = \frac{\sin(70°)}{59.65}$ or $\cos(APB) = \frac{40^2 + 59.65^2 - 60^2}{2(40)(59.65)}$ **[M1]**

Angle $APB = 70.9°$ $(70.93...°)$ **[A1]**

Bearing $= 050 + 70.9 = 120.9°$ **[A1]**

[Back to Question] *[Total Marks: 8]*

22 (a) Bearing of P from $Q = 052° + 180° = 232°$ **[M1]**

Angle $PQR = 232° - 160° = 72°$ **[A1]**

(b) Uses Cosine Rule: $PR^2 = 45^2 + 30^2 - 2(45)(30)\cos(72°)$ **[M1]**

$PR^2 = 2925 - 2700(0.3090...)$ or 2090.65... **[M1]**

45.7 (accept 45.72 to 45.73) **[A1]**

(c) Uses Sine Rule to find $\angle QPR$: $\frac{\sin(\angle QPR)}{30} = \frac{\sin(72°)}{45.72}$ **[M1]**

$\sin(\angle QPR) = 0.624...$ **[M1]**

$\angle QPR = 38.6°$ (or 38.61°) **[A1]**

Bearing $= 052° + 38.6° = 090.6$ **[A1]**

[Back to Question] *[Total Marks: 9]*

23 (a) Identifies angle $PQR = 230° - 120° = 110°$ (or uses parallel lines to find $180 - 50 = 130$ and $360 - 130 - 120 = 110$) **[B1]**

Uses Cosine rule: $PR^2 = 45^2 + 60^2 - 2(45)(60)\cos(110°)$ [M2] (M1 for correct substitution into formula)

$PR = 86.4$ or $86.44...$ **[A1]**

(b) Uses Sine rule: $\frac{\sin(QPR)}{60} = \frac{\sin(110°)}{86.44}$ **[M1]**

Angle $QPR = 40.7°$ $(40.73...)$ **[A1]**

Bearing $= 050° + 40.7° = 090.7°$ **[A1]**

(c) $090.7 + 180$ **[M1]**

270.7 **[A1]**

[Back to Question] *[Total Marks: 9]*

24 **(a)** 270
270 **[B1]**

(b) 018.6
+
180
018.6
+180
[M1]

198.6
198.6 **[A1]**

(c) **(i)** 198.6
48.4
198.6
48.4
[M1]

150.2
150.2 **[A1]**

(ii) Angle
=
90
18.6
=
71.4
ADC=90
18.6
=71.4
[B1]

Angle
=
322.8
270
=
52.8
BCD=322.8
270
=52.8
[B1]

Angle
=
360
(
71.4
+
150.2

+
52.8
)
=
85.6
ABC=360
(71.4
+150.2
+52.8
)=85.6 **[A1]**

(d) Bearing of
A from
=
048.4
+
180
=
228.4
B=048.4
+180
=228.4

[M1]

Bearing of
C from
=
228.4
85.6
=
142.8
B=228.4
85.6
=142.8 **[A1]**

[Back to Question] *[Total Marks: 10]*